Question 1 Planning (15 marks)

Defining the problem (2 marks)

P θ is the independent variable and a is the dependent variable, or vary θ and measure a. [1]

P Keep F constant. [1]

Methods of data collection (4 marks)

M Diagram showing inclined plane with labelled support (not if a ruler used as the inclined plane or as vertical support). [1]

M Method to measure angle e.g. use a protractor to measure θ or use a ruler to measure marked distances from which $\sin\theta$ or θ may be determined. (Allow a labelled protractor in the correct position.) [1]

M Method to measure a time or velocity to determine a, e.g. measure the time using a stopwatch, light gate(s) connected to a timer, motion sensor connected to a time display. [1]

M Use a balance to measure the mass of the trolley. [1]

Method of analysis (3 marks)

A Plot a graph of a against $\sin\theta$. or Plot a graph of ma against $\sin\theta$. or Plot a graph of ma against $mg\sin\theta$. [1]

A Relationship is valid if the graph is a straight line and does **not** pass through the origin [1]

A $k = F - m \times$ (y-intercept) or $k = F -$ (y-intercept) or $k = F -$ (y-intercept) [1]

Do not allow lg-lg graphs.

Additional detail (6 marks)

Relevant points might include: [6]

1 Keep mass of trolley constant/use same trolley.

2 Correct trigonometry relationship to determine $\sin\theta$ or θ using marked lengths.

3 Use ruler to measure appropriate distance to determine a, e.g. length of slope, length of card for light gate method, position of motion sensor.

4 Equation to determine a from measurements taken appropriately with a as the subject.

5 Measurement of F for a valid method e.g. take reading from newton-meter or from stretched elastic/spring from extension (allow falling weight e.g. $F = mg$).

6 Use a constant extension to produce a constant force when using stretched spring/elastic.

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7 Method to ensure the inclined plane is the same height each side of the plane or spirit level across plane or ensure force F (or string) is parallel to the plane.

8 Safety precaution linked to falling mass/trolley or spring/elastic breaking (not string).

9 Rearrangement of relationship into $y = mx + c$ e.g. $ma = -mg \sin\theta + (F - k)$ or $a = -g\sin\theta + \frac{F-k}{m}$ or correct y-intercept (subject must be y-axis).

10 Repeat experiment for each angle θ to find average for a.

Do not allow vague computer methods.

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